

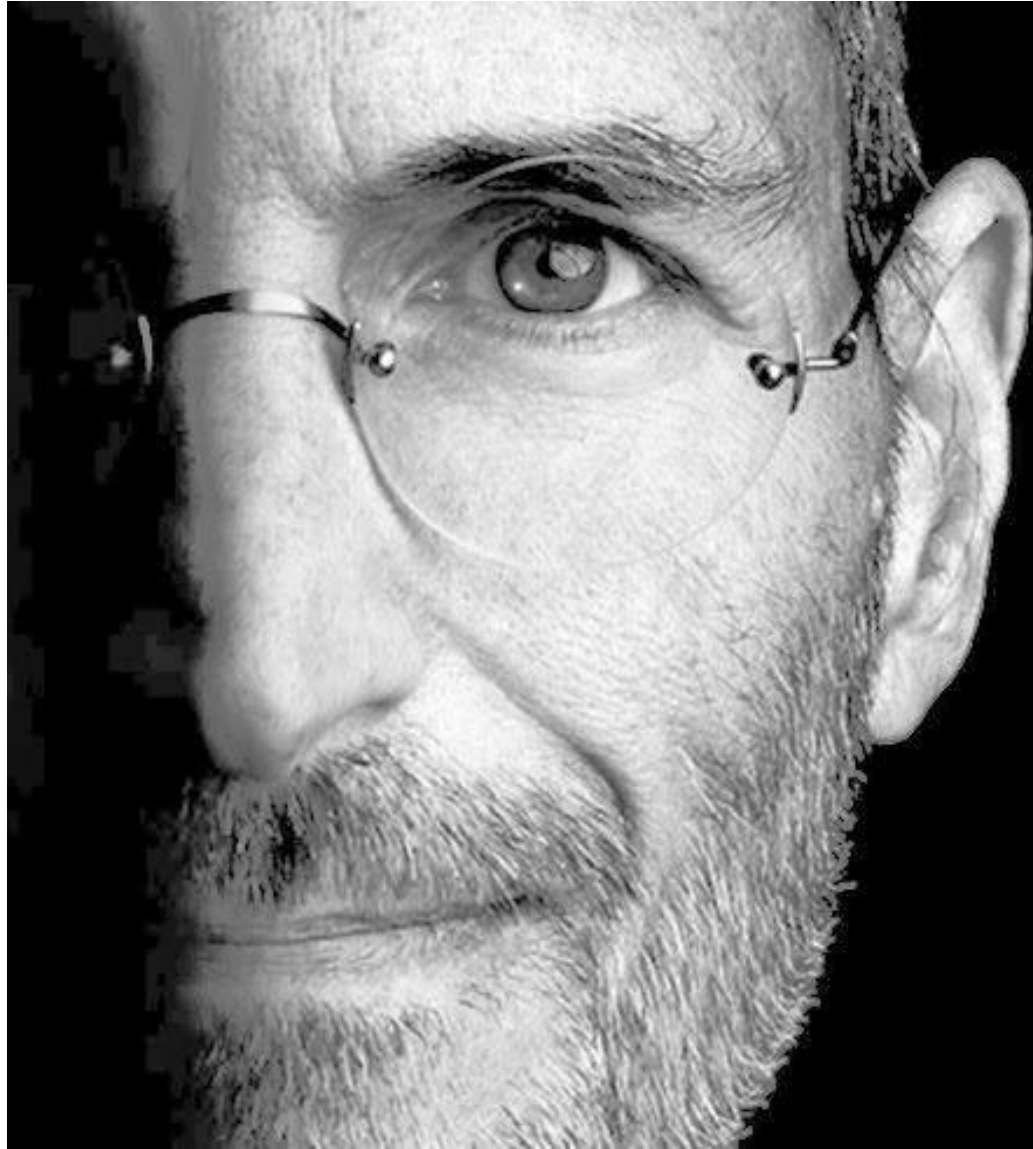
Academic Writing Class 1 AW Group 7

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Centre for Languages and Academic Skills





"You can have the greatest idea in the world,
but if you can't communicate your ideas, it
doesn't matter. "

-Steve Jobs

Need help with your writing skills?

Book a writing consultation

What? Free one-on-one support

Where? On campus or online

When? Sign up for a time slot

For more information

www.tudelft.nl/writingcentre



Today's programme

- Introduction
- Course information
- Introduction scientific writing
- Literature review paper
- Structuring literature reviews
- Steps in the literature review process and the use of AI
- What's next?

Introduction

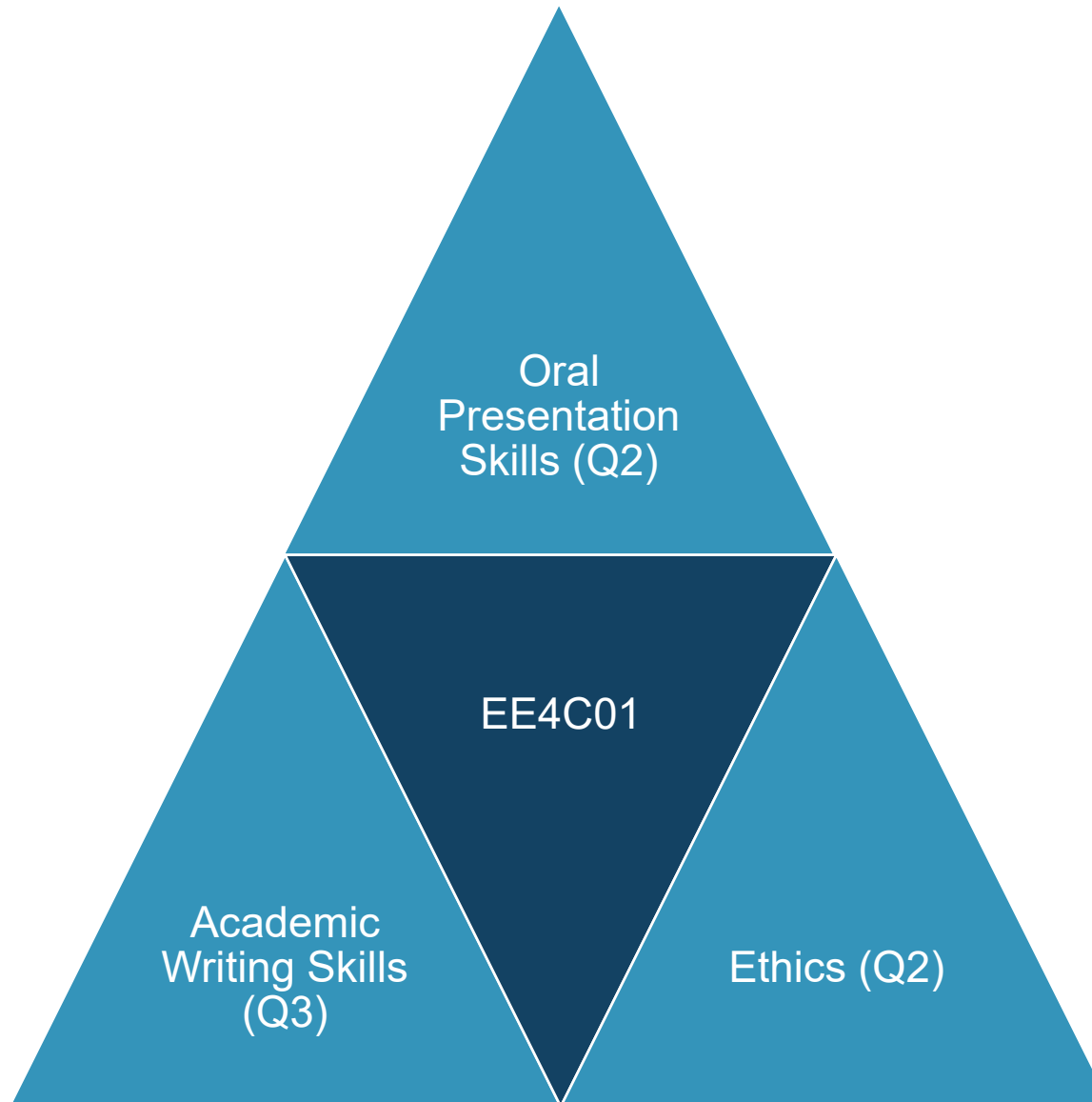
Introduction exercise

In groups of 2 or 3, discuss:

- What is the most difficult text you ever had to write?
- What is the most valuable writing advice you ever got?
- How do you usually approach writing assignments?
- What would you like to learn/improve in this course?

Course Information

EE4C01 Profile Orientation & Academic Skills



Learning objectives

In this course, you will learn how to write a **scientific paper**. To achieve this, you will learn:

- to tailor your text to the purpose, target audience and genre conventions of the text
- to formulate a clear research question
- to structure your text clearly
- to support your claims with clear and convincing arguments
- to critically incorporate literature into your text
- to support results with figures and tables
- to use a scientific writing style
- how you can use generative AI responsibly during the writing process

Course programme

Week	Meeting	Deadline (always 23:59h)
3.1	-	-
3.2	17 Feb: Class 1	20 Feb: Assignment 1 – Research proposal
3.3		27 Feb: Assignment 2 – Introduction & Method
3.4	3 Mar: Class 2	
3.5		11 Mar: Assignment 3 – Body part
3.6		16 Mar: Assignment 4 – Peer feedback
	17 Mar: Class 3	
3.7		25 Mar: Assignment 5 – Draft paper
3.8	31 Mar: Consultation sessions	
3.9		9 Apr: Assignment 6 - Final paper
3.10	-	-

Course rules

- Attendance is mandatory for all meetings
- Be prepared:
 - Submit assignments on time
 - Do the readings

Writing assignment

In teams of 3, write a literature review paper about a topic related to your field of study (more specifically, your track)

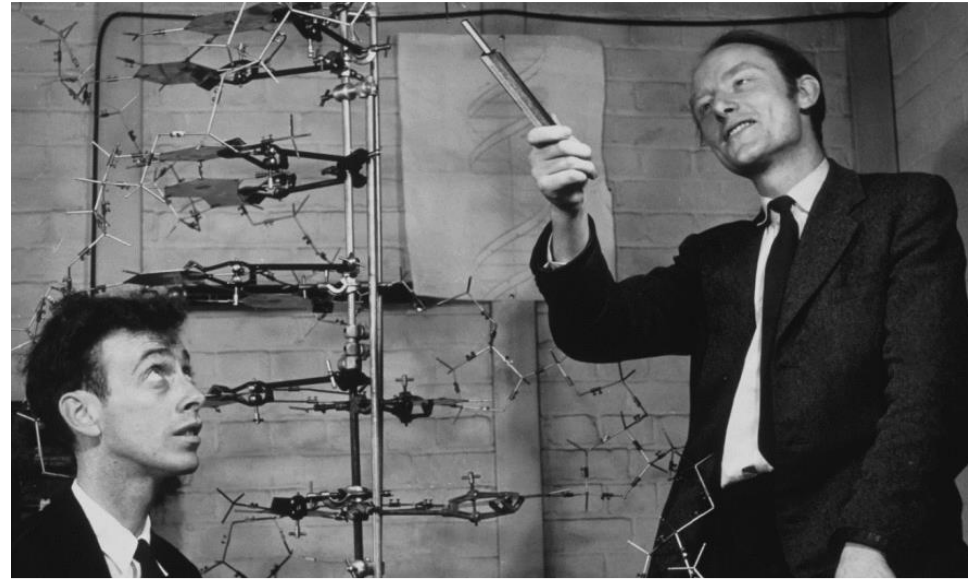
- **Goal:** Provide readers with up-to-date information about a particular topic in your field and add something new
- **Target audience:** peers (MSc EE in general)
- **Word limit:** 2000
- **Sources:** at least 9 academic sources (at least 2 from 2024 or later)
- **Reference style:** IEEE

Introduction Scientific Writing

Rhetoric can make a difference



Oswald Avery



James Watson & Francis Crick

Writing About Biology: How Rhetorical Choices Can Influence the Impact of a Scientific Paper

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ABSTRACT: A comparison of the landmark papers by Watson and Crick (1953) and Avery *et al.* (1944) shows how different rhetorical choices can influence the impact of a scientific paper. The paper by Watson and Crick revolutionized biology and remains a model of scientific rhetoric. In contrast, the paper by Avery *et al.* reports equally important findings, but its impact was greatly diminished because of rhetorical choices made by its authors. Together, these papers provide an

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey¹. They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons:

(1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram). We have made the usual chemical assumptions, namely, that each chain consists of phosphate diester groups joining β -D-deoxy-ribofuranose residues with 3',5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow righthanded helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions.



Each chain loosely resembles Furberg's² model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's standard configuration³, the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3.4 Å. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 Å. The distance of a phosphorus atom from the fibre axis is 10 Å. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so

that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations) it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain, does not appear to be restricted in any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain, is given, then the sequence on the other chain is automatically determined.

It has been found experimentally^{4,5} that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid.

It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data^{5,6} on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in time following, communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in building it, together with a set of co-ordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on interatomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their co-workers at King's College, London. One of us (J.D.W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

J.D. WATSON
F.H. C. CRICK

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

¹Pauling, L., and Corey, R. B. *Nature*, 171, 346 (1953); *Proc. U.S. Nat. Acad. Sci.*, 39, 84 (1953).

²Furberg, S., *Acta Chem. Scand.*, 6, 634 (1952).

³Chargaff, E., for references see Zamenhof, S., Braverman, G., and Chargaff, E., *Biochim. et Biophys. Acta*, 9, 402 (1952).

⁴Wyatt, G.R., *J. Gen. Physiol.*, 36, 201 (1952).

⁵Astbury, W.T., *Symp. Soc. Exp. Biol.*, 1, *Nucleic Acid*, 66 (Camb. Univ. Press, 1947).

⁶Wilkins, M. H. F., and Randall, J. T., *Biochim. et Biophys. Acta*, 10, 102 (1953).

STUDIES ON THE CHEMICAL NATURE OF THE SUBSTANCE INDUCING TRANSFORMATION OF PNEUMOCOCCAL TYPES

INDUCTION OF TRANSFORMATION BY A DESOXYRIBONUCLEIC ACID FRACTION ISOLATED FROM PNEUMOCOCCUS TYPE III

By OSWALD T. AVERY, M.D., COLIN M. MACLEOD, M.D., AND
MACLYN McCARTY,* M.D.

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PLATE 1

(Received for publication, November 1, 1943)

Biologists have long attempted by chemical means to induce in higher organisms predictable and specific changes which thereafter could be transmitted in series as hereditary characters. Among microorganisms the most striking example of inheritable and specific alterations in cell structure and function that can be experimentally induced and are reproducible under well defined and adequately controlled conditions is the transformation of specific types of *Pneumococcus*. This phenomenon was first described by Griffith (1) who succeeded in transforming an attenuated and non-encapsulated (R) variant derived from one specific type into fully encapsulated and virulent (S) cells of a heterologous specific type. A typical instance will suffice to illustrate the techniques originally used and serve to indicate the wide variety of transformations that are possible within the limits of this bacterial species.

Griffith found that mice injected subcutaneously with a small amount of a living R culture derived from *Pneumococcus* Type II together with a large inoculum of heat-killed Type III (S) cells frequently succumbed to infection, and that the heart's blood of these animals yielded Type III pneumococci in pure culture. The fact that the R strain was avirulent and incapable by itself of causing fatal bacteremia and the additional fact that the heated suspension of Type III cells contained no viable organisms brought convincing evidence that the R forms growing under these conditions had newly acquired the capsular structure and biological specificity of Type III pneumococci.

The original observations of Griffith were later confirmed by Neufeld and Levinthal (2), and by Baurhenn (3) abroad, and by Dawson (4) in this laboratory. Subsequently Dawson and Sia (5) succeeded in inducing transformation *in vitro*. This they accomplished by growing R cells in a fluid medium containing anti-R serum and heat-killed encapsulated S cells. They showed that in the test tube as in the animal body transformation can be selectively induced, depending on the type specificity of the S cells used in the reaction system. Later, Alloway (6) was able to cause

Different rhetorical choices: Watson & Crick vs. Avery et al.

	Watson & Crick	Avery et al.
Conciseness	900 words	7500 words
Level of detail	Accessible and easy to read, even for non-experts	Dense and difficult to read, even for experts
Confidence of authors	Confident style ("this is correct")	Hesitant style ("this might be the case")
Clarity of purpose	Thesis mentioned in opening sentence	No thesis in opening paragraph, DNA only mentioned halfway through article
Scientific gap / importance	Importance work stressed ("...novel features which are of considerable interest.")	Description of their work as a "more detailed analysis"

(Moore, 2000)

What is scientific writing?

“There are just three immutable characteristics of good scientific writing that distinguish it from all other literature. It must always be:

- precise
- clear
- brief

... and in that order. If it is vague, it is not scientific writing; if it is unclear or ambiguous, it is not scientific writing and if it long winded and unnecessarily discursive, it is poor scientific writing.”

(Lindsay, 2011, p. 12)

What is a scientific text?

A scientific text is a text that communicates scientific research

It communicates:

- what your main question is (**objective**)
- why that is an interesting question (**motive**)
- how you found the answer (your **method**)
- what your answer is (your **results: support + conclusion**)
- what your answer means (**implications**)
- how definite your answer is (**limitations** to your research)

Literature Review Paper

Each genre has rules



What scientific genres do you know?

- Thesis (bachelor or master, PhD)
- Journal article/research article
- Monography (book about one subject by one author)
- Research proposal
- **Literature review paper**
- Book review
- Annotated bibliography

What is a literature review paper?

Discuss in groups of 2 or 3:

- What are the “rules” of a literature review paper?
- How is it different from other scientific genres (e.g., experimental paper, thesis)?

What is a literature review paper?

“...a literature review **summarizes and evaluates** a body of writings about a specific topic.” (Knopf, 2006, p. 127)

A literature review paper is “...a journal paper that provides a **comprehensive overview** of (or a selection of) the literature in a specific area, bringing together the material in a clearly structured way, and **adding value** through coming to some **interesting conclusions**.” (Van Wee & Bannister, 2016, p. 279)

What is synthesising?

Synthesising simply means combining

- ✗ summarising the main points of each source
- ✓ putting together the ideas and findings of multiple sources in order to make an overall point

How? Look for similarities and differences between your sources

Your synthesis should show the reader where the sources overlap and where they diverge

Bad example (just summarising, no added value)

Franz (2008) studied undergraduate online students. He looked at 17 females and 18 males and found that none of them liked APA. According to Franz, the evidence suggested that all students are reluctant to learn citation styles. Perez (2010) also studied undergraduate students. She looked at 42 females and 50 males and found that males were significantly more inclined to use citation software ($p < .05$). Findings suggest that females might graduate sooner. Goldstein (2012) looked at British undergraduates. Among a sample of 50, all females, all were confident in their abilities to cite and were eager to write their dissertations.

Good example (synthesis)

Claim

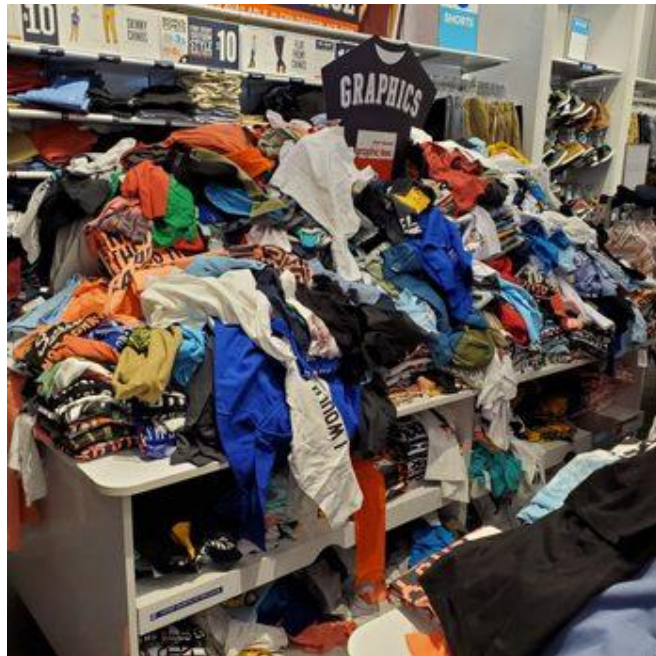
Studies of undergraduate students reveal conflicting conclusions regarding relationships between advanced scholarly study and citation efficacy. Although Franz (2008) found that no participants enjoyed learning a citation style, Goldstein (2012) determined in a larger study that all participants felt comfortable citing sources, suggesting that variables among participant and control group populations must be examined more closely. Perez (2010) expanded on Franz's original study with a larger, more diverse sample, which showed that there is an inconsistency in the lack of joy of learning a citation style and the use of citation software. Although the majority of the students did not enjoy learning a citation style, they did use citation software, although male students tend to use it more often than female students.

Support

What is a literature review paper?

- In a literature review, you answer a **(main) question** based on what was written about it in other studies (instead of your own data, as in an experimental research paper)
- It is not just an overview but a **critical review (comparing and contrasting)**
- There are different types of literature review papers:
 - Systematic literature review (not realistic for you)
 - **Explorative** literature review: describe the state-of-the-art and establish important scientific gaps
 - **Comparative** literature review: compare several methods, technologies, theories, etc.

Structuring Literature Reviews



Standard structure for scientific texts

IMRD

- Introduction
- Method
- Results
- Discussion

Introduction

The introduction consists of three parts:

- **Background/motivation:** background of the topic, problem, what we already know, what is still unknown (gap in literature), relevance to fill the gap
- **Objective:** the objective (aim/purpose/main question/research question), the method used to reach this objective
- **Outline:** organisation of the paper (only if not IMRD)

Introduction: Motivation and objective

Motivation:

- What is the **problem** in society or science that research could help to solve?
- What has **already been researched** to find a solution for this problem?
- What do we **not know yet**, to be able to solve this problem (the missing piece, often called the **scientific gap**)

The motivation leads to the **objective** (research question):

- The aim of this article...

Method

- Your review needs to be replicable
- What steps did you take in searching and selecting sources:
 - Search strategies: e.g., keyword search, snowballing
 - Databases
 - Keywords and Boolean operators (AND, OR, NOT)
 - Selection criteria: e.g., language, time frame, context
 - Quality check, e.g., peer-reviewed journals, number of citations



Results

- Give a full account of the results or your research
- Don't forget the compare and contrast aspect (don't just summarise)
- Think about visualising the results (e.g., a table with an overview)

Discussion (and Conclusion)

- Repeat the research question/aim
- Answer the research question (= **conclusion**)
- Give arguments to support the answer (= **summary** of the results)
- Interpret the conclusion: what does this mean/what is the added value (in relation to existing literature)? (= **implications**)
- Critically evaluate your own research (= **limitations**)
- Give suggestions for future research

Writing Groups

Make writing groups

- Make writing groups of 3 students, preferable from the same track
- Write your names on the list

Steps in the Literature Review Process and the Use of AI

Steps in the literature review process



Using AI or not?

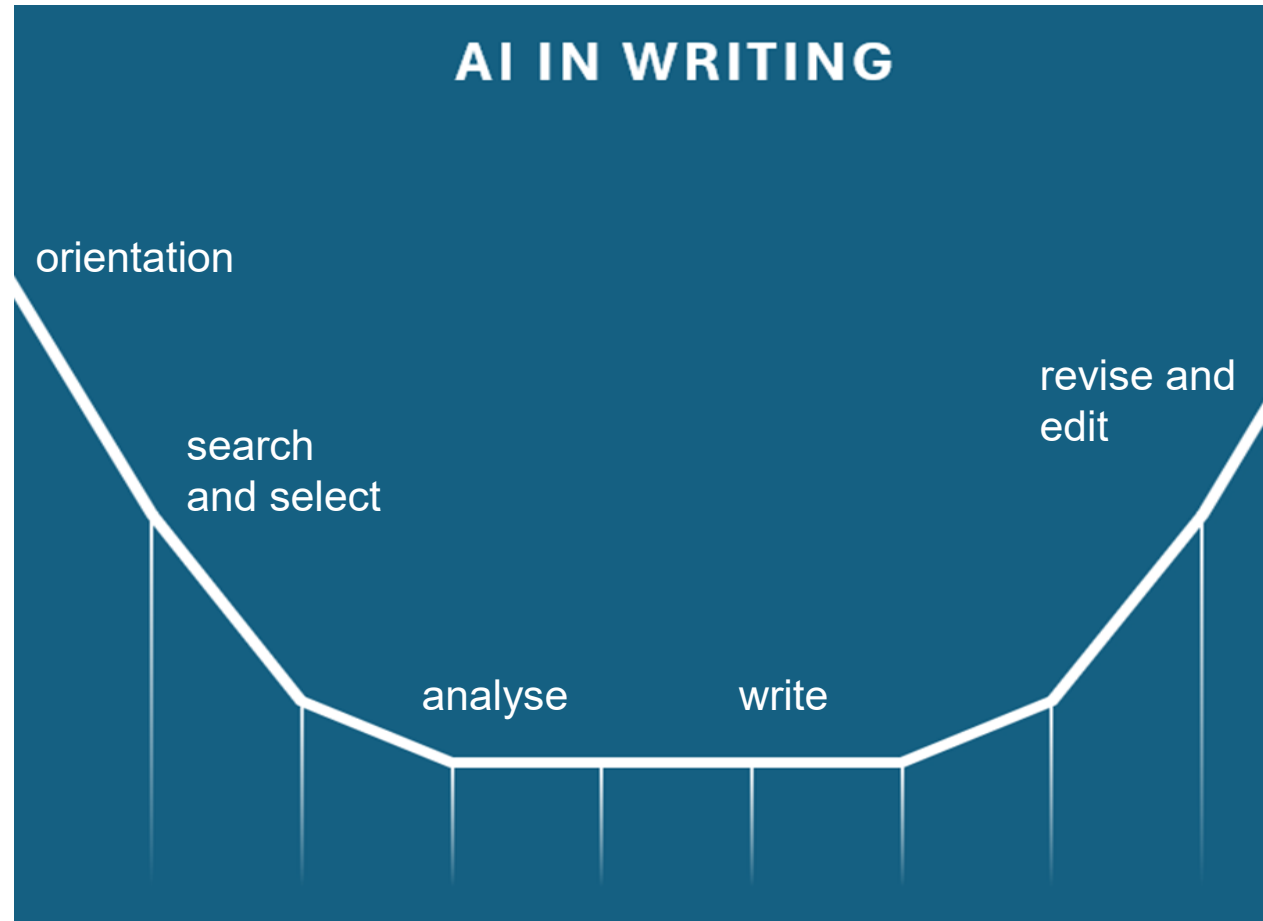
Reasons to use AI

- Makes your life easier
- Can perform tasks quite fast
- If used well, can improve the quality of your work

Reasons not to use AI

- Hallucinations
- Biases
- Cognitive offloading
- Environmental impact
- And so on...

Responsible and effective AI use



General vs academic AI tools

General



Copilot



ChatGPT



Claude



Academic



Elicit



SCISPACE



CONNECTED
PAPERS



ResearchRabbit



Undermind

∠ Litmaps

Responsible and effective AI use

- **Prompt:** Effective GenAI use starts with an effective prompt
- **Proof:** Critically evaluate the quality of the output, using your own critical thinking skills
- **Privacy** (and other considerations): Never feed personal details or confidential information to GenAI tools and also consider the environmental impact of GenAI before using it
- **Presentation:** Don't forget about your own voice when you are writing a text
- **Property:** In the end, always remember that you are responsible for your work, also when you use GenAI

First steps

- Brainstorm on your topic
- Explore academic literature
- Develop research question



First steps

- **Brainstorm on your topic**
- Explore academic literature
- Develop research question



Brainstorm on your topic

- What interests you?
- What are 'hot topics' in your field?
- Examples from previous courses:

**Twinkling Artifact in Doppler Ultrasound, a Useful Tool for
Detecting Kidney Stones**

Different methods for estimating the solar energy
potential in urban environments

Viability assessment of different types of amplifiers
for large scale ECG acquisition

**Graph-Based Image Processing: a Review of
Reducing Computational Complexity**

Exploring State of the art MIMO cellular antennas
for mm-wave 5G communication

Exercise: Brainstorm on your topic

- Suggested AI tool: Copilot or Perplexity
 - If you don't want to use AI, use 'traditional' search engines like Google
- Ask for suggestions for a topic for your literature review paper
 - Example prompt: *You are a master student in [study]. You have to write a literature review paper about [general topic]. Suggest five possible research directions, each with a short description (3-4 sentences) explaining why the topic is relevant and what kinds of research questions could be explored. Present the suggestions as a numbered list.*
- Evaluate the output
 - Which ideas seem relevant, surprising or promising?
 - Which ideas do you want to explore further?

First steps

- Brainstorm on your topic
- **Explore academic literature**
- Develop research question



Exercise: Explore academic literature

- Suggested academic AI tool: Elicit or Perplexity
 - If you don't want to use AI, use 'traditional' search engines like Google Scholar
- Search for literature on your (preliminary) topic
- Evaluate the results
 - Are the sources relevant and trustworthy? (How do you know?)
 - Can you use the sources for further searching (e.g., snowballing)?

Further searching

- In addition to scientific AI tools, use “traditional” search methods, such as Google Scholar, Scopus, Web of Science
- TULib (TU Delft Library) offers a lot of relevant information about finding literature
- TU Delft gives you access to a lot of databases

First steps

- Brainstorm on your topic
- Explore academic literature
- **Develop research question**



Guidelines for research questions

The research question should:

- be neither too broad, nor too narrow
- be clear: it should not contain ambiguous words
- not be too long
- be one question only
- start with a wh-word: what, why, when, who, where, how, to which extent...
- cover everything that is discussed in the text
- lead to an academic debate
- achievable (available time/means)

Remember

- (Gen) AI tools are there to support you
- Do not outsource the process to AI
- Be transparent about your use of AI



Guidelines for Artificial Intelligence (AI)-Generated Text

The use of content generated by artificial intelligence (AI) in an article (including but not limited to text, figures, images, and code) shall be disclosed in the acknowledgments section of any article submitted to an IEEE publication. The AI system used shall be identified, and specific sections of the article that use AI-generated content shall be identified and accompanied by a brief explanation regarding the level at which the AI system was used to generate the content.

The use of AI systems for editing and grammar enhancement is common practice and, as such, is generally outside the intent of the above policy. In this case, disclosure as noted above is recommended.

Statement on the use of generative AI

This article was written within the course EE4C01 Profile Orientation & Academic Writing. Total word count: ##### {here, enter the number of words of the parts that count towards the maximum word count}.

When writing this text, we: * {tick what applies, multiple options are possible if you used generative AI}:

- ☐ did not use generative AI.
- ☐ used generative AI to generate (parts of) the content of the text. Where applicable, we have justified this in a footnote and/or with an in-text citation.
- ☐ used generative AI for inspiration and/or to structure the text.
- ☐ used generative AI to improve the grammar, style and/or spelling of the text.

Summary

Main take aways from class 1

- Assignment: write a literature review
 - 2000 words
 - min. 9 academic sources, min. 2 from 2024 or later
- A literature review is not just summary, but a critical evaluation and synthesis of scientific sources
- AI: use it as a tool, not as the holy grail

What's next?

Course programme

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3.7		25 Mar: Assignment 5 – Draft paper
3.8	31 Mar: Consultation sessions	
3.9		9 Apr: Assignment 6 - Final paper
3.10	-	-

Assignment 1: Research proposal

- Go to Brightspace for the assignment description
- Use the reading materials
- **Deadline:** Friday, 20 February, 23:59
- **Feedback:** Teacher feedback via Brightspace

Assignment 2: Introduction and method

- Go to Brightspace for the assignment description
- Use the reading materials
- **Deadline:** Friday, 27 February, 25:59h
- **Feedback:** Peer feedback in class 2